Spiking Reservoir State-Space Encoding for Perturbation-Characterized Affective EEG Signal Analysis

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***Abstract*—Affective electroencephalographic (EEG) event-related potentials (ERPs) are noisy, subject-dependent, and only partially observed projections of neural dynamics, and endpoint accuracy alone does not reveal what temporal information a representation retains or discards. We define a fixed leaky integrate-and-fire (LIF) reservoir with a six-bin spike-count (BSC_6) code as a temporal encoding operator, compress it for a linear readout, and audit it under subject-grouped validation against classical band-power and ERP-window baselines under amplitude-noise, temporal-jitter, and channel-dropout perturbations. Clean classification favors ERP-window features (balanced accuracy 0.616, vs. 0.485 band-power and 0.463 reservoir), but the reservoir embedding shows a distinct stability profile: it retains 97–99% of its clean balanced accuracy under 5 dB amplitude noise and 30% channel dropout, versus 74–84% for the baselines, so the clean-performance gap narrows sharply under controlled feature-space corruption. Controls indicate this stability is not a dimensionality artifact (a dimensionality-matched random projection of the baseline degrades like the baseline) and that all encodings exceed a label-permutation null; a single-bin spike-count ablation locates the effect in the fixed reservoir state-space expansion rather than temporal bin granularity.**

***Index Terms*—EEG, event-related potentials, spiking neural networks, reservoir computing, biomedical signal processing, affective computing, perturbation analysis.**

I. Introduction

Electroencephalographic event-related potentials (ERPs) are time-locked measurements of neural response dynamics that offer millisecond-scale temporal information for biomedical data analysis, yet the observed waveform is a noisy, low-dimensional, and subject-dependent projection of an unobserved neural dynamical system [12], [13]. Inter-subject variability, trial averaging, volume conduction, and measurement noise can cause endpoint classifiers to conflate genuine temporal structure with subject-specific nuisance variation, so a representation that performs well on one clean split may not preserve the same signal properties when amplitude noise, temporal misalignment, or channel loss is introduced.

Reporting only endpoint accuracy does not reveal *which* temporal features a representation preserves or suppresses. This gap is acute for spiking encodings: fixed recurrent reservoirs and spiking networks are expressive nonlinear temporal expansions [1], [4], but in EEG/ERP work they are most often evaluated as end-to-end classifiers [9], [15] rather than characterized as temporal operators whose discrete spike-count observables remain traceable to post-stimulus time windows.

The perturbation behavior of affective-ERP reservoir encodings—their response to amplitude noise, temporal jitter, and channel dropout—is rarely reported, even though preprocessing and acquisition choices materially affect EEG representations [16]–[18].

This paper studies a fixed spiking reservoir state-space encoding for affective ERP signal analysis and audits it as a temporal encoding operator (Fig. 1). Each ERP observation is mapped through a fixed leaky integrate-and-fire (LIF) reservoir, summarized by six post-onset spike-count bins (BSC_6), and compressed for a linear readout. The paper isolates this reservoir temporal-encoding component; graph topology, structure–function coupling, and closed-loop analyses belong to other parts of a broader doctoral framework and are outside this submission. The contributions are:

1) A fixed LIF/BSC_6 temporal encoding operator for affective EEG/ERP observations, defined as a composition of a nonlinear reservoir map, spike-count binning, and an unsupervised projection (Sec. III, Eq. (4)).
2) Evaluation of the reservoir embedding against classical band-power and ERP-window baselines under subject-grouped validation, using balanced accuracy, macro-F1, and macro one-vs-rest AUC with per-fold confidence intervals and a permutation null (Sec. V-A, Table II).
3) Perturbation-response characterization under amplitude noise, temporal jitter, and channel dropout, separating representation-level corruption from raw-signal recomputation (Sec. V-B, Table IV, Fig. 3).
4) Controls that bound common confounds—fold-local PCA, a dimensionality-matched random projection, a label-permutation null, and a single-bin spike-count ablation—and identify the fixed reservoir embedding, rather than temporal bin granularity, as the operative factor (Sec. V-A, Table III).

Clinical labels and affective categories define the external task structure; the reservoir-based temporal representation is the object of study.

II. Related Work

ERP and EEG analysis have traditionally relied on time-window amplitudes and latencies, time-domain descriptors such as Hjorth parameters [10], spectral and wavelet summaries [11], and linear statistical models [13]. These features

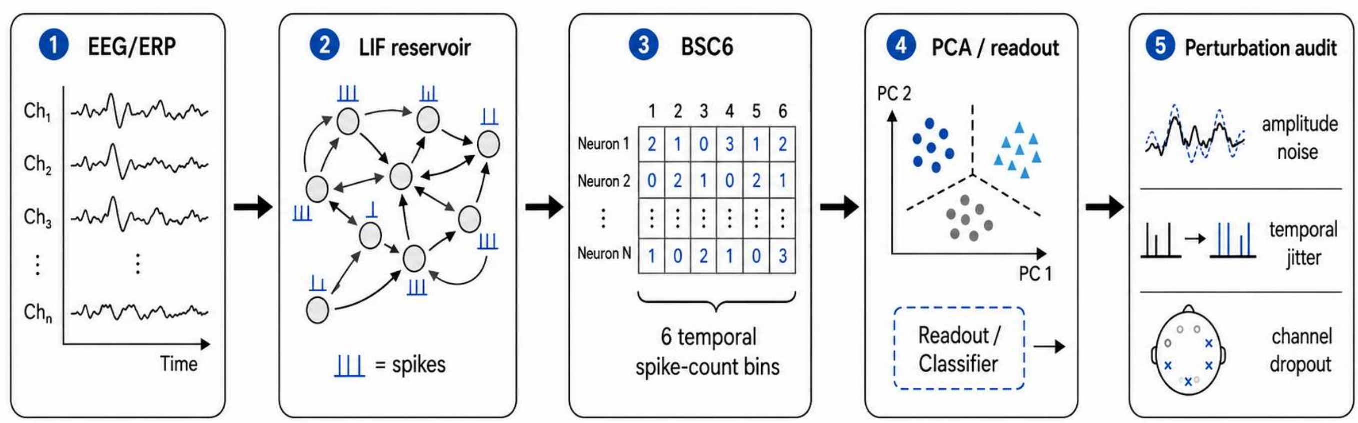

Fig. 1. Reservoir encoding and perturbation-audit pipeline. Trial-averaged affective EEG/ERP observations are mapped through a fixed LIF reservoir, summarized by six temporal spike-count bins (BSC_6), compressed for linear readout, and evaluated under amplitude noise, temporal jitter, and channel-dropout perturbations.

are interpretable and correspond to known components such as the P300 and late positive potential, and source-level analyses show that evoked responses carry structured temporal dynamics [12]. Fixed handcrafted windows may, however, underrepresent nonlinear temporal interactions and can be sensitive to subject-specific variability.

Reservoir computing offers an alternative in which a high-dimensional recurrent dynamical system transforms an input sequence into a separable state trajectory [1]–[4]. In liquid state machines, recurrent spiking dynamics induce a nonlinear temporal basis on which a simple readout is trained; related spiking models encode information in spike timing and counts [5], [6]. This separation of fixed dynamics from a trained readout is attractive for biomedical signal analysis because the reservoir can be studied as a measurement operator rather than as a fully optimized black-box classifier.

Spiking and deep networks have been applied to EEG and ERP signals [9], [14], [15], but most studies emphasize end-point accuracy, while preprocessing and acquisition choices that strongly affect EEG representations are often under-examined [18]. For perturbation-sensitive physiological data, accuracy alone is insufficient: a representation should also be assessed through subject-level generalization, controlled corruption, and representation-geometry analysis [19]. Prior work in the ARSPI-Net line developed neuromorphic reservoir feature extraction for affective EEG [7], [8]; the present paper isolates and audits the spiking temporal-encoding component as a measurement operator.

III. Methods

A. ERP Observation Model

Let $X_s^{(c)} \in \mathbb{R}^{C\times T}$ denote the trial-averaged ERP observation for subject s and affective condition c, with C electrodes (channels) and T temporal samples; observations are stored as $T\times C$ arrays and transposed before encoding. We treat $X_s^{(c)}$ as a noisy observation of an unobserved neural state trajectory:

$$\xi_{t+1} = f_\theta(\xi_t, u_t) + \eta_t, \qquad x_t = H\xi_t + \epsilon_t, \tag{1}$$

where ξ_t is the latent neural state, H is a measurement operator induced by scalp observation, and ϵ_t captures measurement and preprocessing noise. The perturbations below act directly on this observation model (Sec. V-B).

B. Spiking Reservoir Encoding

For each electrode trace, the signal is passed through a fixed LIF reservoir of 256 neurons with recurrent weights W scaled to spectral radius $\rho = 0.9$ (near the edge of stability), input weights W_{in}, leak parameter $\beta = 0.05$, and firing threshold ϑ. A compact discrete-time representation is

$$v_{t+1} = \beta v_t + W z_t + W_{\text{in}} x_t - \vartheta r_t, \tag{2}$$

$$z_{t+1} = \mathbf{1}[v_{t+1} \ge \vartheta], \tag{3}$$

where v_t is the membrane state, z_t is the binary spike vector, and r_t denotes reset after threshold crossing. The reservoir is fixed during feature extraction; only the downstream readout is trained.

Post-onset activity is summarized by six equal temporal bins over the window $t \in [10, 70]$ in sample units, which at 256 Hz spans approximately 39–273 ms post-stimulus. For reservoir unit j and bin k, the binned spike-count code is $b_{j,k} = \sum_{t\in\mathcal{B}_k} z_j(t)$, $k = 1,\ldots,6$. Per-electrode spike-bin features are compressed to 64 principal components and concatenated across the 34 channels into a trial-level reservoir representation $E_s^{(c)} \in \mathbb{R}^{2176}$. Writing $s_t = z_t \in \{0,1\}^N$ for the reservoir spike state, the encoding is a fixed composition of a nonlinear temporal map, spike-count binning, and the unsupervised projection:

$$E = \phi_{\text{PCA}}\big([b_1,\ldots,b_6]\big) = \Phi(X), \tag{4}$$

so that $E = \Phi(X)$ is a fixed, label-free observable of the input. The principal-component projection is estimated once, unsupervised and without affective labels, over the spike-bin vectors pooled across all observations and electrodes; because the basis is fitted on every subject—including those later held out—it is transductive with respect to subject inclusion, and we report it as a fixed representation audit rather than leakage-free preprocessing (bounded by the fold-local control in Sec. V-A). As a temporal-structure control, we also form a single-bin code (BSC_1, the total post-onset spike count per unit) processed through the identical PCA pipeline. This window is an early, compact interval that excludes the later P300/LPP ranges available to the ERP-window baseline (Sec. III); the clean-classification comparison is therefore conservative for the reservoir.

Fig. 2 shows the reservoir-derived computational objects for the three affective conditions: the LIF spike raster, the membrane-potential response, and the BSC_6 embedding projected to two principal components.

C. Baselines and Readout

Two classical baselines are evaluated under the same subject-grouped protocol. The band-power baseline (A0) uses spectral band-power features (5 bands $\times$ 34 channels $=$ 170 dimensions). The ERP-window baseline ($\mathrm{A0}_e$) uses mean amplitude in three post-stimulus windows—early (80–200 ms), P300 (250–450 ms), and late positive potential (450–800 ms)—per channel ($3 \times 34 = 102$ dimensions). To separate representational robustness from raw dimensionality, we add a dimensionality-matched control: A0 randomly projected to 2176 dimensions (Gaussian random projection, averaged over five random seeds). A balanced L2-regularized logistic readout is used throughout to test representational separability without a high-capacity classifier.

D. Perturbation Diagnostics

Two perturbation regimes are separated. Representation-level perturbations corrupt the extracted feature vector; raw-signal recomputation applies perturbations before feature extraction and recomputes the reservoir representation, propagating corruption through reservoir dynamics. Formally, a perturbation $\mathcal{P}_\eta$ acts either on the representation, $E \mapsto \mathcal{P}_\eta(E)$, or on the observation model of Eq. (1): amplitude noise augments the measurement noise ϵ_t, channel dropout zeroes rows of the measurement operator H, and temporal jitter shifts the sampling index of x_t. We apply additive amplitude noise (signal-to-noise ratios), temporal jitter of the alignment window, and channel dropout at increasing electrode-removal rates. Performance is balanced accuracy (BA) and macro-F1.

IV. Experimental Protocol

A. Dataset and Task

The experiments use a trial-averaged affective ERP dataset with 211 subjects after exclusion of one anomalous subject record. Each subject contributes three affective conditions (negative, neutral, pleasant), yielding 633 subject-condition observations. The ERP traces contain 34 scalp channels downsampled to 256 Hz, with 256 post-stimulus samples per epoch, baseline-corrected relative to onset. Subject-level data are maintained in a restricted research environment; only aggregate, de-identified results are reported. Table I summarizes the protocol.

TABLE I
DATASET AND EVALUATION PROTOCOL SUMMARY.

Item	Value
Subjects (after exclusion)	211
Observations	633 (subject $\times$ condition)
Conditions	3 (neg., neutral, pleasant)
Channels / rate / samples	34 / 256 Hz / 256
Baseline (A0) dim.	170 (5$\times$34)
Reservoir embedding (E) dim.	2176 (64$\times$34)
Validation	subject-grouped 5-fold CV

TABLE II
CLEAN SUBJECT-LEVEL PERFORMANCE (MEAN $\pm$ 95% CI OVER 25 FOLDS). PERMUTATION-NULL BA $= 0.338 \pm 0.022$.

Cfg	Features	BA	F1	AUC
$\mathrm{A0}_e$	ERP-window (102)	0.616 ± 0.016	0.615	0.790
A0	Band-power (170)	0.485 ± 0.015	0.483	0.665
A1	Reservoir BSC_6 (2176)	0.463 ± 0.013	0.460	0.644
–	Reservoir BSC_1 (2176)	0.467 ± 0.011	0.465	0.646

B. Validation

All results are computed under a single protocol: StratifiedGroupKFold (5 folds, seeds 42–46), train-only standardization, and a balanced L2 logistic readout; observations from the same subject never span train and test. We report means with 95% confidence intervals (CIs) over the 25 folds, and macro one-vs-rest AUC from pooled out-of-fold probabilities. Chance is calibrated by a label-permutation null (100 permutations).

V. Results

A. Clean Classification and Controls

Table II reports clean three-class performance. The ERP-window baseline ($\mathrm{A0}_e$) gives the strongest clean separability (BA 0.616), above band-power (A0, 0.485) and the reservoir embedding (A1, 0.463), which does not exceed either baseline. A label-permutation null gives BA 0.338 ± 0.022, so every configuration encodes genuine signal (all exceed the null by more than five standard deviations) while the absolute clean separability remains modest.

Temporal-binning ablation. The single-bin total-count code (BSC_1) matches the six-bin code (BSC_6) on clean classification (0.467 vs. 0.463; Table II) and under perturbation (Sec. V-B). We therefore treat BSC_6 as a temporally localized *audit scaffold* that keeps the observable traceable to post-stimulus windows; the ablation shows that both separability and robustness are driven by the fixed reservoir state-space expansion, not by bin granularity.

Transduction and component controls. A fold-local PCA refit (per-fold training subjects) leaves performance unchanged (Table III); varying the component count over $\{16, 32, 64, 128\}$ keeps BA within 0.461–0.491, so neither the transductive basis nor the 64-component choice inflates results.

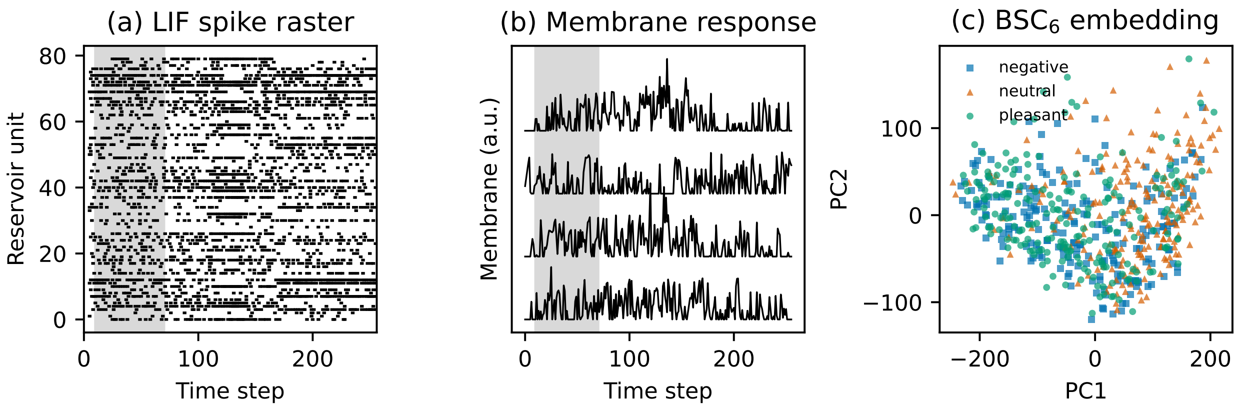

Fig. 2. Reservoir-derived computational objects for affective ERP observations. (a) LIF spike raster for an example trace (shaded region = the $t \in [10, 70]$ analysis window). (b) Membrane response for four units (offset vertically). (c) BSC_6 embedding projected to the first two principal components, colored by affective condition.

TABLE III

FOLD-LOCAL VS. GLOBAL PCA FOR THE RESERVOIR EMBEDDING E (A1). OVR-AUC IS MACRO ONE-VS-REST FROM POOLED OUT-OF-FOLD PROBABILITIES.

PCA basis	BA	Macro-F1	OvR-AUC
Global pooled (transductive)	0.463	0.460	0.644
Fold-local (per-fold train)	0.465	0.462	0.649

B. Representation-Level Perturbation Response

Under feature-space corruption (Table IV) the ranking inverts. The ERP-window baseline, strongest on clean data, is the most fragile, retaining only 75% and 74% of its clean BA at 5 dB noise and 30% dropout; band-power retains 81–84%; the reservoir embedding retains 97–99% ($0.463 \rightarrow 0.449$ and 0.456). The clean-performance gap thus narrows sharply under controlled corruption. Across the 25 folds, the reservoir's mean absolute BA drop is 0.013 at 5 dB noise and 0.007 at 30% dropout, versus 0.155 and 0.162 for ERP-window and 0.090 and 0.077 for band-power features; the paired per-fold degradation is significantly smaller for the reservoir than for either baseline under both perturbations (Wilcoxon signed-rank, $p < 0.001$). Two controls show the reservoir's stability is not an artifact of representation size. The dimensionality-matched random projection of A0 (2176-D, five seeds) gives clean BA 0.501 ± 0.002 but collapses to 0.399 ± 0.008 at 5 dB, tracking A0 (0.395) rather than E (0.449); and channel dropout is channel-matched (all use 34 channels). The single-bin BSC_1 embedding is equally robust (0.452 at 5 dB, 0.454 at 30% dropout), confirming the binning plays no role.

C. Raw-Signal Recomputed Perturbations

A stricter test corrupts the signal before feature extraction and recomputes the embedding under the same protocol (Table V). Clean recomputation reproduces the embedding result (0.465). Temporal jitter is the most damaging perturbation ($0.465 \rightarrow 0.422$), with amplitude noise and channel dropout intermediate (0.449, 0.431); all remain above the permutation null. The embedding is thus reasonably stable even when corruption enters before extraction, but temporal alignment is the dominant sensitivity, consistent with the dependence of the

TABLE IV

REPRESENTATION-LEVEL PERTURBATION: BALANCED ACCURACY WITH RETENTION (PERTURBED/CLEAN) IN PARENTHESES. A0 RANDOM-PROJ. IS THE DIMENSIONALITY-MATCHED CONTROL (MEAN OVER FIVE SEEDS).

Representation	Clean	Amp. 5 dB	Drop 30%
ERP-window (102)	0.616	0.461 (75%)	0.454 (74%)
Band-power A0 (170)	0.485	0.395 (81%)	0.408 (84%)
A0 random-proj. (2176)	0.501	0.399 (80%)	–
Reservoir E (2176)	0.463	0.449 (97%)	0.456 (99%)

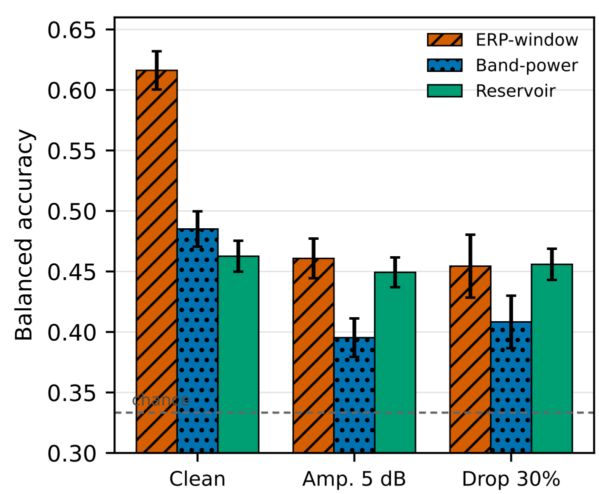

Fig. 3. Representation-level perturbation response. Balanced accuracy (mean, 95% CI) for the ERP-window and band-power baselines and the reservoir embedding under clean, 5 dB amplitude-noise, and 30% channel-dropout conditions. The ERP-window baseline leads on clean data but degrades most; the reservoir embedding shows the smallest absolute degradation, narrowing the gap under controlled corruption. Dashed line = chance.

TABLE V
RAW-SIGNAL RECOMPUTATION DIAGNOSTICS (MEAN $\pm$ 95% CI OVER 25 FOLDS).

Condition	BA	Macro-F1
Clean recomputation	0.465 ± 0.013	0.462
Temporal jitter, ± 50 ms	0.422 ± 0.016	0.379
Amplitude noise, 5 dB	0.449 ± 0.018	0.445
Channel dropout, 30%	0.431 ± 0.021	0.402

reservoir state trajectory on the input timing.

VI. DISCUSSION

The evidence supports a bounded reading. The fixed reservoir embedding encodes genuine affective-condition signal (well above the permutation null) but does not match the classical baselines on clean static classification: ERP-window amplitude features are clearly strongest (0.616), and band-power (0.485) also exceeds the reservoir (0.463). The reservoir's distinguishing property is stability: under representation-level amplitude noise and channel dropout it retains 97–99% of clean accuracy while the baselines lose 16–26%, so the strongest clean representation (ERP-window) becomes among the weakest under corruption and the clean-performance gap narrows sharply. Two controls establish that this stability is not a measurement artifact—a dimensionality-matched random projection of band-power degrades like band-power, and the dropout comparison is channel-matched. Under raw-signal recomputation the embedding remains reasonably robust, with temporal jitter the dominant vulnerability, reflecting the reservoir's dependence on input timing.

A key negative result delimits the contribution: a single-bin total spike count (BSC_1) matches the six-bin code on clean classification and under every perturbation. Temporal bin granularity is therefore not the operative factor; the robustness is a property of the fixed reservoir state-space embedding and its unsupervised compression, not of the temporal binning. The audit framing—characterizing what a fixed encoding preserves under controlled corruption—is what the data support, rather than a claim that temporal structure per se confers an advantage.

Several scope boundaries remain. The dataset is a single affective ERP cohort, and external validation is needed before generalization. The comparators here are classical; a modern deep baseline (e.g., EEGNet) under the same subject-grouped perturbation protocol is the most important next step. The study evaluates software-level representations, not neuromorphic hardware energy or latency, which would require measurement on dedicated platforms [20], [21]. The perturbation diagnostics are controlled tests, not substitutes for full acquisition artifacts, and graph topology and structure–function coupling are not evaluated here.

VII. CONCLUSION

A fixed spiking-reservoir state-space embedding of affective ERPs encodes genuine but modest condition signal and does not beat classical band-power or ERP-window baselines on clean classification. Its distinguishing property is perturbation stability: it retains 97–99% of clean balanced accuracy under representation-level amplitude noise and channel dropout, where the baselines lose far more, narrowing the clean-performance gap under corruption; dimensionality-matched and channel-matched controls show this is not an artifact of representation size. A single-bin spike-count ablation shows the stability comes from the reservoir state-space expansion rather than temporal bin granularity, and the embedding's chief vulnerability is temporal misalignment at acquisition. These results characterize the operating regime of a fixed reservoir encoding for affective ERP analysis. The most important next step is a modern deep baseline (e.g., EEGNet) under the same perturbation protocol; multi-cohort generalization and alternative spiking codes are further directions.

Data Availability and Scope. The EEG/ERP data used in this study are access-controlled research data and are not redistributed with this submission. Analyses are reported only as aggregate, de-identified results. Clinical labels, where present in the source cohort, are used only to define biomedical signal-analysis context and are not interpreted as diagnostic decisions.

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Research Experience

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Bioinformatics Analyst, Feinstein Institute for Medical Research 2011–2014

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Mentoring and Service

- Faculty advisor, Upsilon Pi Epsilon Computer Science Honor Society, St. Joseph's University.
- Undergraduate research mentor for projects in neural machine translation, visual speech recognition, affective computing, and AI bias.

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Neuromorphic computing; spiking neural networks; liquid-state machines; reservoir computing; graph neural networks; EEG/ERP signal processing; perturbation analysis; dimensionality reduction; Python; machine learning; LaTeX.